

SPEECH UNDERSTANDING AS A TRAINING SIGNAL FOR GENERATION: SHARED SUPERVISION AND LATENT RECONSTRUCTION FEEDBACK

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ABSTRACT

Speech generation must faithfully convey the intended text while maintaining natural acoustic quality. Auxiliary speech-to-text prediction can transfer linguistic knowledge through shared parameters, but supervision on real audio does not directly evaluate generated content. We propose a training framework that combines shared understanding supervision with direct content feedback in latent space. Shared learning retains auxiliary text prediction during generation-focused training. We further introduce Reconstruct CE, which differentially reconstructs audio latents from the flow-training state and applies transcript supervision through a frozen copy of the understanding branch. This connects local velocity regression with endpoint content supervision, without waveform decoding in the reconstruction branch or changes to generation-time decoding. Matched continuation experiments show that generation-emphasized shared learning reduces Chinese CER and English WER by 10.7% and 10.4% relative to generation-only continuation. In a separate matched-update ablation, Reconstruct CE reduces these errors by 6.4% and 8.0% relative to shared supervision, while improving predicted naturalness and maintaining or improving speaker similarity at the reported precision. The converged model achieves the highest speaker similarity in both languages among the evaluated systems under their respective generation settings. Audio samples are available online.

Index Terms— speech generation, understanding supervision, flow matching, text reconstruction, multi-task learning

1. INTRODUCTION

As speech generation tasks become more complex, synthesis must jointly ensure faithful text realization, naturalness, and speaker preservation. This motivates moving beyond direct text-to-acoustic mapping toward understanding-informed generation. By learning the reverse audio-to-text relationship, speech understanding captures the correspondence between linguistic content and acoustic realization, providing complementary knowledge to guide synthesis.

Shared speech-text modeling offers one route for transferring this knowledge. SpeechT5 [1] learns shared representations through encoder-decoder pretraining. ViOLA [2] unifies speech recognition, synthesis, and translation through autoregressive codec language modeling, while UniVoice [3] combines autoregressive recognition

with flow-based synthesis. Broader audio language models extend joint learning across understanding and generation tasks [4, 5]. An auxiliary transcription objective evaluated only on real audio transfers supervision through shared parameters, but does not itself assess the linguistic content of reconstructed speech.

Feedback from synthesized audio provides a complementary route. The machine speech chain [6] reconstructs text from synthesized speech, StyleTTS 2 [7] uses pretrained speech-model representations for adversarial supervision, and Cont-SPT [8] applies ASR supervision to reconstructed audio during tokenizer pretraining. These studies demonstrate the value of speech-analysis feedback beyond auxiliary learning on real audio.

For flow-based generation, such feedback must connect local acoustic predictions with endpoint content. Flow matching [9], used in speech and audio generators such as F5-TTS [10] and MMAudio [11], supervises local recovery velocities, whereas transcript supervision evaluates whether the reconstructed endpoint expresses the intended text. Differentiable reconstruction can bridge these objectives and propagate content gradients to the generator. This motivates our central question: **How can knowledge learned from speech understanding be effectively transferred to improve speech generation?**

We investigate this question through shared text prediction on real audio and Reconstruct CE. The latter applies transcript supervision to differentially reconstructed speech latents through a frozen understanding branch. Matched bilingual experiments support their complementary benefits for content accuracy and predicted naturalness while preserving speaker similarity at the reported precision. Comparisons with six re-evaluated systems show that the converged model achieves the highest speaker similarity in both languages under the evaluated settings.

2. UNDERSTANDING SUPERVISION FOR GENERATION

Our framework combines shared understanding supervision with reconstructed-content feedback through a common latent interface (Fig. 1).

2.1. Problem Formulation and Backbone

Let x_1 , y , and a denote clean audio latents, transcript, and caption. Generation conditions on $c = [a, \text{SEP}, y]$ and available audio context. Our 1.69B-parameter MMDiT has 18 joint blocks, width 1024, and 64-dimensional latents. The tasks share the backbone, embeddings,

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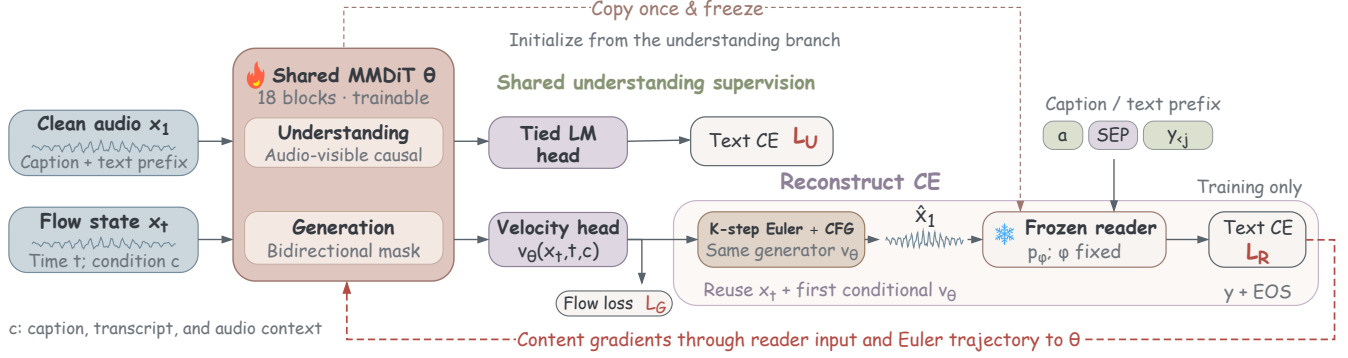


Fig. 1. Understanding supervision for speech generation. Shared learning combines flow matching with text prediction on real audio. Reconstruct CE reuses the flow-training state and a first-step conditional velocity prediction for K differentiable CFG-guided Euler steps. Transcript gradients pass through a frozen reader’s input and the trajectory to the generator. Reconstruction is training-only. a/y : caption/transcript; waveforms represent audio latents.

and projections, with a tied text output head. Following UniVoice [3], generation uses bidirectional audio–text attention; understanding restricts audio to audio and text to audio plus preceding text, preventing future-text leakage.

2.2. Shared Understanding Supervision

For Gaussian noise ϵ and $t \sim \mathcal{U}(0, 1)$, flow matching uses $x_t = tx_1 + (1 - t)\epsilon$. The generation loss $\mathcal{L}_G$ is the mean squared error between $v_\theta(x_t, t, c)$ and $x_1 - \epsilon$, averaged over valid predicted frames and latent dimensions. Known audio context is fixed to clean latents and excluded from this loss.

Shared text prediction trains the generation backbone to recover linguistic content, retaining audio-to-text supervision as generation receives greater weight. At $t = 1$, the model predicts text from clean audio. The understanding loss $\mathcal{L}_U$ is teacher-forced cross-entropy, with caption, separator, transcript, and EOS tokens as targets. We combine the objectives as $\mathcal{L}_{\text{joint}}^{(r)} = (r\mathcal{L}_G + \mathcal{L}_U)/(r + 1)$. We compare fixed and generation-emphasized ratios r ; reconstruction starts from a separate joint checkpoint (Sec. 3.1).

2.3. Differentiable Latent Reconstruction

Reconstruct CE extends the sampled flow-training state to a differentiable endpoint. For reconstruction, we combine conditional and unconditional velocity predictions using classifier-free guidance (CFG):

$$v_\theta^{\text{cfg}}(z, t, c) = (1 + g_{\text{rec}})v_\theta(z, t, c) - g_{\text{rec}}v_\theta(M_\emptyset z, t, c_\emptyset), \quad (1)$$

where g_{rec} is the reconstruction guidance strength. The masking operator $M_\emptyset$ zeros only known audio-context positions in z , preserving the region to be generated. The unconditional input $c_\emptyset$ zeros all text embeddings (including caption and transcript) and any separate reference-audio latents; lengths and masks are retained.

With $z_0 = x_t$ and $h = (1 - t)/K$, we obtain $\hat{x}_1 = z_K$ through K differentiable Euler steps:

$$z_{k+1} = \Pi_c \left[z_k + h v_\theta^{\text{cfg}}(z_k, t + kh, c) \right], \quad (2)$$

for $k = 0, \dots, K - 1$. Here, Π_c restores clean known audio context and clears padding at every step.

At the first Euler step, we reuse the conditional velocity prediction from flow training and additionally compute the unconditional prediction to form the guided velocity. Subsequent steps use the same CFG formulation. Gradients propagate through both velocity branches along the reconstruction trajectory.

2.4. Frozen-Reader Content Supervision

We initialize reader p_ϕ once from the same joint checkpoint used to initialize the generator and keep its parameters frozen. Its input gradients update the generator through the Euler trajectory. At $t = 1$, it reads $\hat{x}_1$, including reconstructed and fixed-context regions. Here y is the transcript of the complete audio utterance, so the reader evaluates the full utterance represented jointly by the fixed context and reconstructed region, rather than only the newly generated span:

$$\mathcal{L}_R = -\frac{1}{N_y} \sum_{j \in \mathcal{T}_y} \log p_\phi(y_j | \hat{x}_1, a, y_{<j}), \quad (3)$$

where $\mathcal{T}_y$ contains transcript and EOS positions and N_y counts valid target tokens. Caption and separator tokens provide context. Samples with dropped text conditioning contribute to flow matching but are excluded from $\mathcal{L}_R$.

Reconstruction training optimizes

$$\mathcal{L}_{\text{recon}} = \frac{10}{12}\mathcal{L}_G + \frac{1}{12}\mathcal{L}_U + \frac{1}{12}\mathcal{L}_R. \quad (4)$$

CE-off removes only $\mathcal{L}_R/12$, keeping the other coefficients fixed; the unfrozen-reader ablation also updates ϕ . The reconstruction branch needs no waveform decoding and runs only during training, leaving generation inference unchanged.

3. EXPERIMENTAL SETUP

3.1. Data and Controlled Training

The training mixture has 26.23M entries including replay, with Chinese/English proportions of 60/40. We use caption-plus-transcript conditioning, random-span inpainting, and text-conditioning dropout of 0.2. Seed-TTS-eval [12] contains 2020 Chinese and 1088 English cases. Our controlled configurations share reference audio, transcripts, and assignments of two fixed captions per language.

Table 1. Seed-TTS-eval. A: Self-evaluated baselines and our reconstruction model; B: matched continuations (300k+200k). All systems are evaluated using the same Seed-TTS evaluation set and the same metric implementations. Chinese uses CER and English uses WER. Acoustic metrics are reported when available. MOS: 20 listeners $\times$ 8 utterances per system/language; arithmetic mean of 160 ratings/cell (Sec. 3.3).

Model / setting	Params (B)	Chinese				English			
		CER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑
<i>A. Self-evaluated baselines and our reconstruction model</i>									
Qwen2.5-Omni-7B	10.73	2.74	0.75	3.53	4.50	5.60	0.63	4.19	4.06
MiMo-Audio-7B-Instruct	8.02	3.10	0.58	2.66	3.06	4.03	0.44	3.54	3.13
Step-Audio 2-mini	8.32	1.41	0.64	3.51	3.88	1.89	0.47	4.31	3.75
UniAudio 2.0	4.86	5.63	0.48	3.09	3.00	4.76	0.34	3.84	3.31
Ming-Lite-Omni	0.69 (Talker only)	2.05	0.61	3.73	3.19	5.62	0.45	4.24	3.75
Ming-UniAudio-16B-A3B	18.27	1.06	0.70	2.77	3.56	1.82	0.58	3.67	3.44
Reconstruct CE (converged)	1.69	1.69	0.77	3.35	3.75	3.02	0.68	3.94	3.88
<i>B. Controlled continuations with shared supervision</i>									
Generation-only continuation	1.69	2.52	0.73	3.12	—	4.23	0.67	3.76	—
Joint 10:1	1.69	2.00	0.74	3.18	—	3.83	0.67	3.80	—
Joint 20:1	1.69	2.25	0.74	3.20	—	3.79	0.68	3.80	—

Table 2. Matched-update reconstruction: Joint 10:1 800k EMA +100k, $K = 10$, reconstruction guidance $g_{\text{rec}} = 4$ for both reader variants, three inference seeds. CE-off removes only $\mathcal{L}_R/12$.

Training setting	Chinese			English		
	CER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$	WER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$
CE-off (shared supervision)	1.88	0.75	3.18	3.39	0.65	3.86
Reconstruct CE (unfrozen reader)	1.86	0.77	3.16	3.52	0.67	3.85
Reconstruct CE (frozen reader)	1.76	0.77	3.27	3.12	0.67	3.90

Two controlled continuations separate shared understanding from reconstruction feedback. Table 1B starts all branches from Joint 10:1 300k and adds 200k updates (500k total). Generation-only continuation removes the understanding objective, Joint 10:1 retains it, and Joint 20:1 increases the generation-to-understanding weight. Table 2 starts all variants from Joint 10:1 800k EMA and adds 100k updates ($K = 10$, $g_{\text{rec}} = 4$; 900k total). CE-off retains the same flow and understanding loss coefficients; reconstruction variants add transcript supervision with a trainable or frozen reader, both using the same reconstruction CFG setting. Table 1A reports the converged frozen-reader model after 300k reconstruction updates from the same 800k initialization (1.1M total), using $K = 10$ and $g_{\text{rec}} = 4$ throughout.

3.2. Baseline Evaluation

We evaluate released Qwen2.5-Omni-7B [13, 14], MiMo-Audio-7B-Instruct [15, 16], Step-Audio-2-mini [17, 18], UniAudio 2.0 [4], Ming-Lite-Omni [19], and Ming-UniAudio-16B-A3B [20]. Qwen uses fixed Chelsie without reference conditioning, transcript-constrained greedy Thinker decoding, and a 4096-token Talker limit. MiMo uses reference-prompt TTS with a voice-matching instruction (global/local temperatures 0.6/0.9; top- p 1.0/0.95). Step uses temperature 0.7, a 4096-token limit, and reference-conditioned token-to-wave decoding. Ming-Lite uses its Talker/zero-shot frontend with reference speech, transcript, and speaker features. UniAudio uses `llm_ep2/ReasoningCodec` and `prompt_instruct_tts`, reference tokens, target text, and “Natural speech with a neutral speaking style.” (temperature 0.9, top- k 50, no CFG, 10 codec steps). Ming-UniAudio uses native text normalization, reference audio/transcript, 10 diffusion iterations, CFG=2, and 16-kHz output.

Baselines produce one output per case, with utterance-specific seeds derived from 42. Ming-Lite has 1086 English outputs after two

decoder failures; all other counts are complete. UniAudio retries one invalid-codec case with another seed, without exclusions. Evaluators are shared, but generation interfaces differ; Qwen’s SIM is not a matched voice-cloning comparison.

Parameter counts exclude standalone codecs/vocoders and our training reader, but include Qwen’s integrated token-to-wave module. Ming-Lite counts only its Talker; MoE counts all experts; tied weights count once.

3.3. Generation and Metrics

Our model uses EMA weights and per-utterance sampling with $v_{\text{cfg}} = (1 + g)v_{\text{cond}} - gv_{\text{uncond}}$, $g = 4$, and zeroed reference in the unconditional branch. Chinese/English use 48/64 Euler steps, time shift 1, and a common VAE posterior-sampling procedure for reference latents. Chinese CER uses Paraformer [21]; English WER uses Whisper large-v3 [22] through *faster-whisper*. SIM uses WavLM-Large features [23] with ECAPA-TDNN [24] (`wavlm_large_finetune.pth`) and cosine similarity to paired reference audio; UTMOS [25] predicts naturalness. Automatic metrics are utterance means. Our Tables 1–2 average three inference seeds per trained checkpoint.

For subjective evaluation, 20 listeners each rated eight utterances per system in each language using a blinded, randomized 1–5 quality scale. Each system–language cell therefore contains 160 valid ratings, and MOS is their arithmetic mean.

3.4. Auxiliary Protocols

Historical curves use learning rate 5×10^{-4} , seed 42, dynamic batches, and 2048 understanding tokens. Generation-only training and Joint 10:1 load compatible weights from one source, but new text parameters may differ; Joint 20:1 restarts at Joint 10:1 300k.

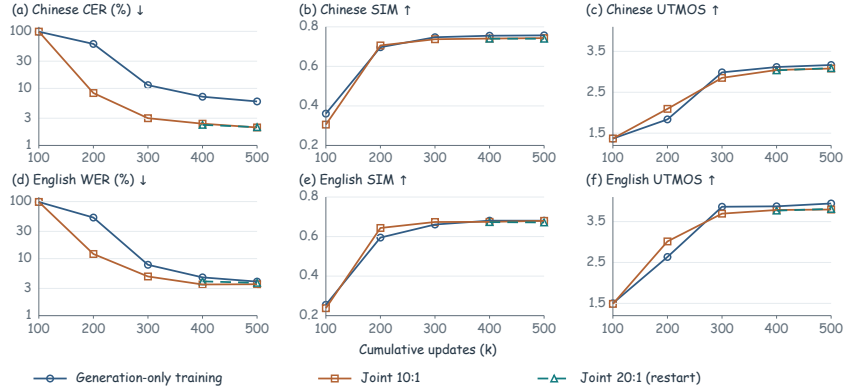


Fig. 2. Historical learning trajectories under the protocol in Sec. 3.4 (seed 42; 100k–500k updates at 100k intervals; logarithmic CER/WER axes). Joint 20:1 starts from Joint 10:1 300k and resets optimizer/data progress; the axis includes initialization updates (Sec. 3).

Table 3. Pilot: Joint 800k EMA, CFG=4 (two branches/step); *: selected pilot sampling depth. Errors (%). ZH/EN: Chinese/English.

Euler steps	4	5	8	10*	12	14	16
ZH CER ↓	3.57	2.65	1.98	1.89	1.89	1.85	1.88
ZH SIM ↑	0.67	0.69	0.73	0.73	0.73	0.73	0.73
ZH UTMOS ↑	2.47	2.95	3.20	3.23	3.24	3.24	3.24
EN WER ↓	3.92	3.58	3.47	3.41	3.29	3.35	3.31
EN SIM ↑	0.60	0.62	0.62	0.63	0.62	0.62	0.62
EN UTMOS ↑	3.49	3.73	3.91	3.92	3.94	3.95	3.94

The historical runs describe learning trajectories; Tables 1B and 2 provide matched comparisons. The pilot uses CFG=4, batches of five, preserved punctuation, and historical reference conditioning. The pilot evaluates inference-time sampling depth, whereas training reconstruction uses $K = 10$ guided Euler steps starting from the sampled flow-training state. Main inference uses 48/64 steps.

4. RESULTS AND DISCUSSION

Shared understanding benefits generation. In Table 1B, Joint 10:1 reduces Chinese CER/English WER from 2.52%/4.23% to 2.00%/3.83% relative to generation-only continuation. Joint 20:1 reaches 2.25%/3.79%, corresponding to relative reductions of 10.7%/10.4%, while increasing UTMOS from 3.12/3.76 to 3.20/3.80. With initialization and updates matched, improvements at both joint-learning ratios support retaining understanding supervision during generation-focused training. The 10:1-to-20:1 change has little, mixed effect; the main gain comes from retaining understanding supervision.

Reconstruction adds endpoint supervision. In Table 2, the frozen-reader variant reduces CER/WER from 1.88%/3.39% to 1.76%/3.12%, yielding 6.4%/8.0% relative reductions versus CE-off. CE-off retains understanding supervision with unchanged flow and understanding loss coefficients. The comparison therefore supports the value of supervising reconstructed content beyond auxiliary prediction on real audio. Flow matching supervises local velocity estimates, while Reconstruct CE evaluates the transcript recoverable from their integrated endpoint. The bilingual improvements are consistent with endpoint feedback supplying a content constraint complementary to the local acoustic objective.

Freezing strengthens content feedback while preserving speech

fidelity. Freezing reduces CER/WER from 1.86%/3.52% to 1.76%/3.12% compared with updating the reader jointly. Freezing keeps the audio-to-text mapping fixed, directing reconstruction feedback to the generator rather than adapting the reader to its outputs. External waveform recognizers measure the accuracy gains, demonstrating benefits beyond the training reader itself. Relative to CE-off, frozen reconstruction also raises UTMOS from 3.18/3.86 to 3.27/3.90 and increases SIM from 0.75/0.65 to 0.77/0.67.

Converged content accuracy and speaker preservation. After 300k reconstruction updates from the 800k joint checkpoint, the model achieves 1.69% CER and 3.02% WER (Table 1A). Its SIM (0.77/0.68) is highest among the evaluated systems, and all three automatic measures improve over MiMo-Audio and UniAudio 2.0 in both languages. Compared with reference-conditioned Ming-UniAudio, it achieves higher SIM (0.77/0.68 versus 0.70/0.58) and UTMOS (3.35/3.94 versus 2.77/3.67). Under the generation interfaces described in Sec. 3.2, these results highlight speaker preservation alongside accurate text realization. The external comparisons characterize the converged system, while the matched continuations isolate the contributions of its supervision design.

Learning trajectories and sampling depth. Historical trajectories (Fig. 2) show earlier error reductions with joint learning, consistent with Table 1B. In the separate pilot (Table 3), 4 to 10 Euler steps reduce CER/WER from 3.57%/3.92% to 1.89%/3.41%, while SIM and UTMOS approach relatively stable ranges around 8–10 steps.

5. CONCLUSION

We presented a training framework that transfers speech understanding to generation through shared learning and latent reconstruction feedback. Shared text prediction retains linguistic supervision during generation-focused training, while Reconstruct CE extends this supervision to reconstructed speech through a frozen understanding branch. The resulting feedback connects local flow-based predictions with endpoint transcript consistency. Controlled bilingual experiments support the complementary benefits of these paths for content accuracy and predicted naturalness while preserving speaker similarity at the reported precision. By operating directly on audio latents, reconstruction requires no waveform decoding, and the reader is used only during training. Together, these findings support understanding supervision as a means of improving speech generation without changing generation-time decoding.

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